W01 – Pandas

Data Mining
Dr Andrea Visentin

Thanks to Dr Alejandro Arbelaez

Python Libraries

Data Science with Python

Get Data

- Beautiful Soup
- LXML
- Tweepy
- **Pandas**

Data Preprocessing

- **Pandas**
- NLTK
- **Scikit**
- **Matplotlib**

Analysis And Modeling

- **Numpy**
- Scipy
- Sympy
- Sklearn
- statsmodels

Evaluate and Present

- iPython
- Flask

Python Libraries for Data Science

- Many popular Python toolboxes/libraries:
  - NumPy
  - SciPy
  - Pandas
  - SciKit-Learn

- Visualization libraries
  - matplotlib
  - Seaborn

and many more …

Python Libraries for Data Science

- NumPy:
  - Introduces objects for multidimensional arrays and matrices, as well as functions that allow to easily perform advanced mathematical and statistical operations on those objects
  - Provides vectorization of mathematical operations on arrays and matrices which significantly improves the performance
  - Many other python libraries are built on NumPy

Link: http://www.numpy.org/

Python Libraries for Data Science

- SciPy:
  - Collection of algorithms for linear algebra, differential equations, numerical integration, optimization, statistics and more
  - Part of SciPy Stack
  - Built on NumPy

Link: https://www.scipy.org/scipylib/

Python Libraries for Data Science

- Pandas:
  - Adds data structures and tools designed to work with table-like data (similar to Series and Data Frames in R)
  - Provides tools for data manipulation: reshaping, merging, sorting, slicing, aggregation etc.
  - Allows handling missing data

Link: http://pandas.pydata.org/

pandas

$y_{it} = \beta' x_{it} + \mu_i + \epsilon_{it}$

Python Libraries for Data Science

- SciKit-Learn:
  - Provides machine learning algorithms: classification, regression, clustering, model validation etc.
  - Built on NumPy, SciPy and matplotlib

Link: http://scikit-learn.org/

Python Libraries for Data Science

- matplotlib:
  - Python 2D plotting library which produces publication quality figures in a variety of hardcopy formats
  - A set of functionalities similar to those of MATLAB
  - Line plots, scatter plots, bar charts, histograms, pie charts etc.
  - Relatively low-level; some effort needed to create advanced visualization

Link: https://matplotlib.org/

Python Libraries for Data Science

- Seaborn:
  - Based on matplotlib
  - Provides high level interface for drawing attractive statistical graphics
  - Similar (in style) to the popular ggplot2 library in R

Link: https://seaborn.pydata.org/

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Pandas

Using Pandas to load CSV or Tables

- Pandas is spreadsheet software for Python
  - A table is called a DataFrame
  - A 1-D array of numbers is called a Series
- Important Features
  - Easily load CSV, TSV files
  - Can easily load data in chunks if needed
  - Support group-by, indexing selecting, merge operations
  - Data Analysis Functions like mean, median

Why pandas?

- One of the most popular library that data scientists use
- Labeled axes to avoid misalignment of data
  - Data[:, 2] represents weight or weight2?
  - When merge two tables, some rows may be different
- Missing values or special values may need to be removed or replaced

	height	Weight	Weight2	age	Gender
Amy	160	125	126	32	2
Bob	170	167	155	-1	1
Chris	168	143	150	28	1
David	190	182	NA	42	1
Ella	175	133	138	23	2
Frank	172	150	148	45	1

	salary	Credit score
Alice	50000	700
Bob	NA	670
Chris	60000	NA
David	-99999	750
Ella	70000	685
Tom	45000	660

Overview

- Created by Wes McKinney in 2008, now maintained by Jeff Reback and many others.
  - Author of one of the textbooks: Python for Data Analysis
- Powerful and productive Python data analysis and Management Library
- Panel Data System
- Its an open source product.

Overview - 2

- Python Library to provide data analysis features similar to: R, MATLAB, SAS
- Rich data structures and functions to make working with data structure fast, easy and expressive.
- It is built on top of NumPy
- Key components provided by Pandas:
  - Series
  - DataFrame

Import

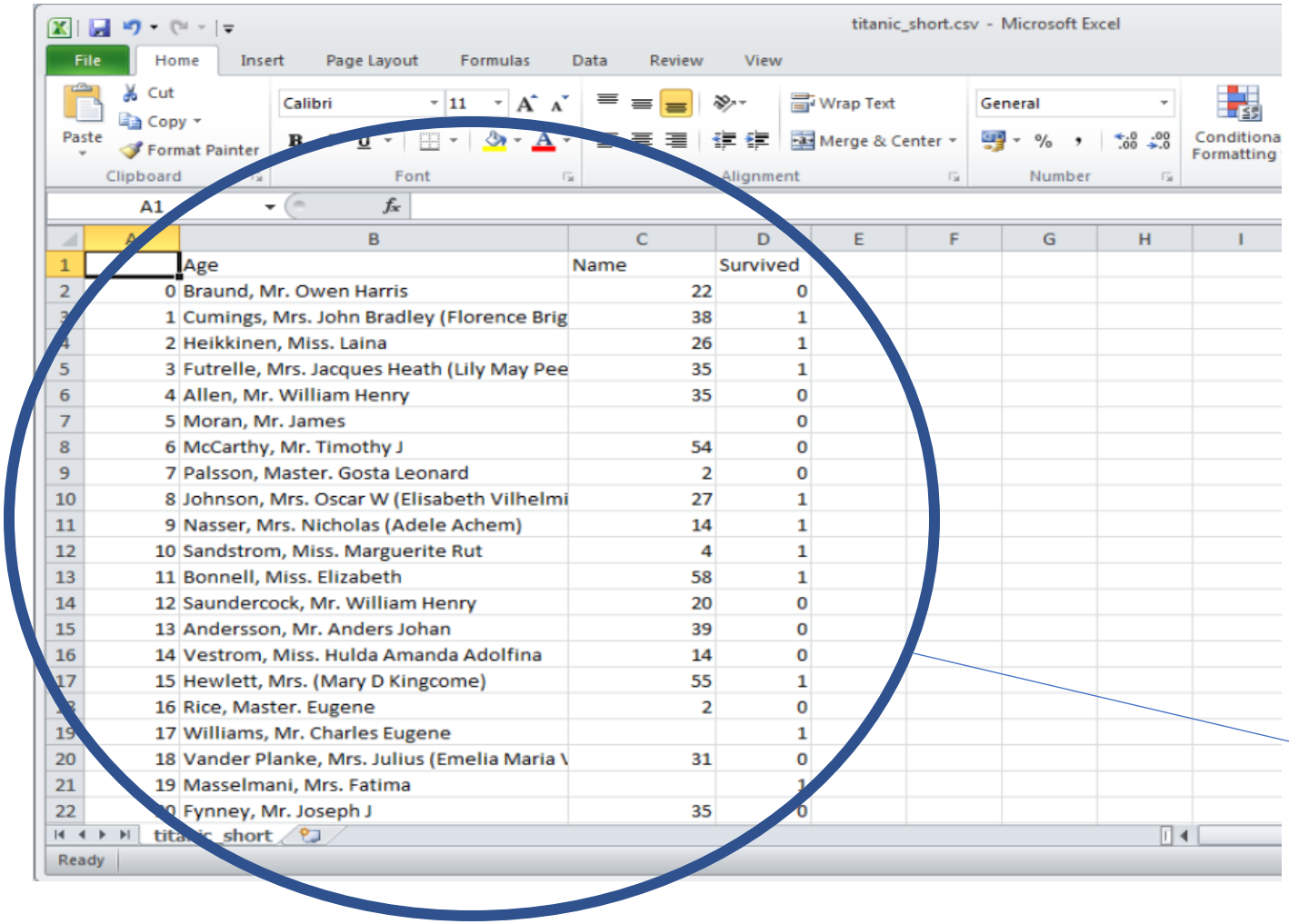
```
In [11]: # import pandas library

         import pandas as pd
```

DataFrame

- A DataFrame represents a tabular, spreadsheet-like data structure containing an ordered collection of columns, each of which can be different value type (numeric, string, Boolean, etc)
- The DataFrame has both row and column index; it can be though of as a dictionary of Series (one for all sharing the same index)

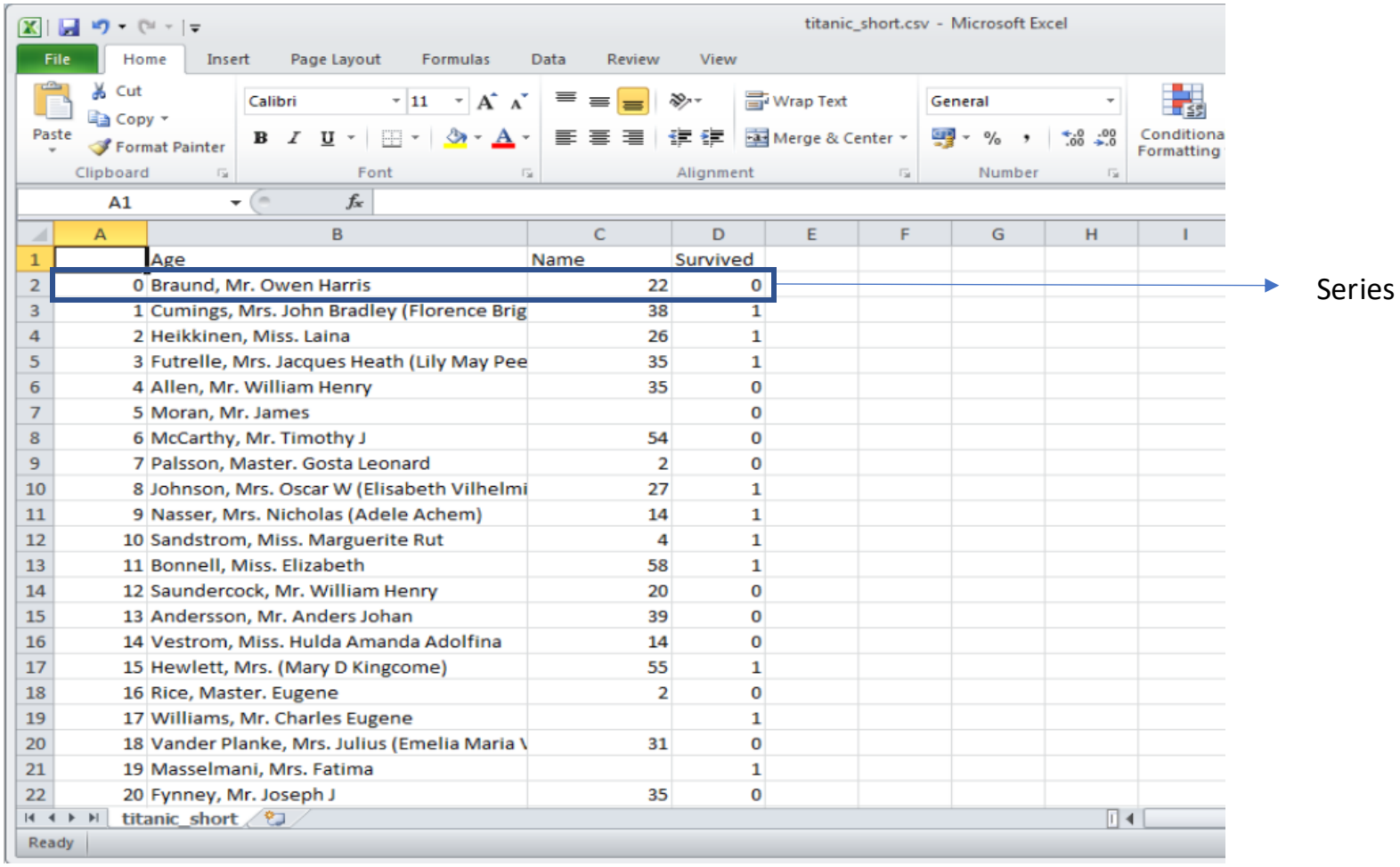
Pandas



	A	B	C	D
1		Age	Name	Survived
2	0	Braund, Mr. Owen Harris	22	0
3	1	Cumings, Mrs. John Bradley (Florence Brig	38	1
4	2	Heikkinen, Miss. Laina	26	1
5	3	Futrelle, Mrs. Jacques Heath (Lily May Pee	35	1
6	4	Allen, Mr. William Henry	35	0
7	5	Moran, Mr. James		0
8	6	McCarthy, Mr. Timothy J	54	0
9	7	Palsson, Master. Gosta Leonard	2	0
10	8	Johnson, Mrs. Oscar W (Elisabeth Vilhelmi	27	1
11	9	Nasser, Mrs. Nicholas (Adele Achem)	14	1
12	10	Sandstrom, Miss. Marguerite Rut	4	1
13	11	Bonnell, Miss. Elizabeth	58	1
14	12	Saundercock, Mr. William Henry	20	0
15	13	Andersson, Mr. Anders Johan	39	0
16	14	Vestrom, Miss. Hulda Amanda Adolfina	14	0
17	15	Hewlett, Mrs. (Mary D Kingcome)	55	1
18	16	Rice, Master. Eugene	2	0
19	17	Williams, Mr. Charles Eugene		1
20	18	Vander Planke, Mrs. Julius (Emelia Maria V	31	0
21	19	Masselmani, Mrs. Fatima		1
22	20	Fynney, Mr. Joseph J	35	0

DataFrame

Pandas



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Series

Data frames

DataFrame

- A DataFrame is a tabular data structure comprised of rows and columns, akin to a spreadsheet or database table.
- It can be treated as an order collection of columns
  - Each column can be a different data type
  - Have both row and column indices

DataFrame

A **DataFrame** is a dictionary of series (dict of series):

```
d = {
      ’names’ : [”bob”, “ken”, “art”, “joe”],
      'grades’ : [50, 90, 100, 45]
}
```

DataFrame - Creation

```
import pandas as pd
data = {
      'state': ['Ohio', 'Ohio', 'Ohio', 'Nevada', 'Nevada'],
      'year': [2000, 2001, 2002, 2001, 2002],
      'pop': [1.5, 1.7, 3.6, 2.4, 2.9]}

frame = pd.DataFrame(data)
print(frame)
```

Reading and Writing Data

CSV files

- A **CSV file** is a file with a ".csv" file extension, e.g. "data.csv", "super_information.csv". The "CSV" in this case lets the computer know that the data contained in the file is in **comma separated value** format, which we'll discuss below.

- CSV is a standard for storing tabular data in text format, where **commas** are used to separate the different columns, and **newlines** (carriage return / press enter) used to separate rows.
  - Typically, the first row in a CSV file contains the names of the columns for the data.

CSV files

Data opened in Microsoft Excel Spreadsheet

	A	B	C	D
1	Name	Age	Country	Goals
2	James Rodriguez	26	Colombia	6
3	Thomas Muller	28	Germany	5
4	Lionel Messi	31	Argentina	4
5	Neymar	26	Brazil	4
6	Robin van Persie	34	Netherlands	4

Column names are first row in text file

Data in CSV format opened in Text Editor

```
1 Name,Age,Country,Goals
2 James Rodriguez,26,Colombia,6
3 Thomas Muller,28,Germany,5
4 Lionel Messi,31,Argentina,4
5 Neymar,26,Brazil,4
6 Robin van Persie,34,Netherlands,4
7
```

Commas are used to separate columns

CSV files

- Note that almost any tabular data can be stored in CSV format – the format is popular because of its simplicity and flexibility.
- You can create a text file in a text editor, save it with a .csv extension, and open that file in Excel or Google Sheets to see the table form.

File Loading

When specifying file names to the read_csv function, you can supply both absolute or relative file paths.

- A **relative path** is the path to the file if you start from your current working directory. In relative paths, typically the file will be in a subdirectory of the working directory and the path will not start with a drive specifier, e.g. (data/test_file.csv).
- An **absolute path** is the complete path from the base of your file system to the file that you want to load, e.g. c:/Documents/Shane/data/test_file.csv. Absolute paths will start with a drive specifier (c:/ in Windows, or '/' in Mac or Linux).

File Loading

We will load files from a github repository:
https://github.com/andvise/DataAnalyticsDatasets/

You can create your own one to store the datasets you want to use. The repository must be public.

We do it to make them available on the internet.

File Loading

main DataAnalyticsDatasets / coffee.csv Go to file ···

andvise Add files via upload Latest commit 2d91808 4 days ago History

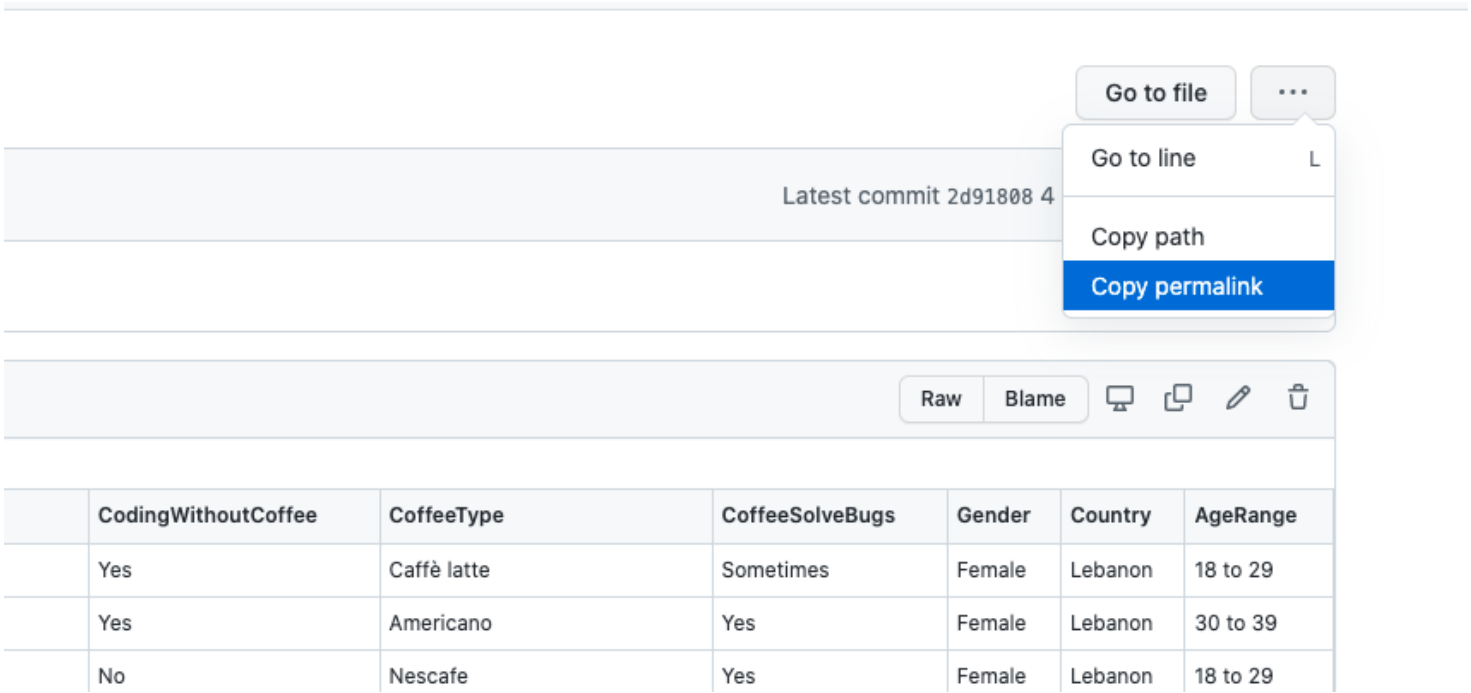
1 contributor

101 lines (101 sloc) | 6.61 KB Raw Blame

Search this file...

	CodingHours	CoffeeCupsPerDay	CoffeeTime	CodingWithoutCoffee	CoffeeType	CoffeeSolveBugs	Gender	Country	AgeRange
2	8	2	Before coding	Yes	Caffè latte	Sometimes	Female	Lebanon	18 to 29
3	3	2	Before coding	Yes	Americano	Yes	Female	Lebanon	30 to 39
4	5	3	While coding	No	Nescafe	Yes	Female	Lebanon	18 to 29
5	8	2	Before coding	No	Nescafe	Yes	Male	Lebanon	NA
6	10	3	While coding	Sometimes	Turkish	No	Male	Lebanon	18 to 29
7	8	2	While coding	Sometimes	Nescafe	Yes	Male	Lebanon	30 to 39
8	5	2	While coding	Yes	Nescafe	Sometimes	Male	Lebanon	NA
9	10	4	Before coding	Sometimes	Turkish	Sometimes	Male	Lebanon	18 to 29
10	10	2	While coding	Yes	American Coffee	Sometimes	Male	Lebanon	30 to 39

File Loading



	CodingWithoutCoffee	CoffeeType	CoffeeSolveBugs	Gender	Country	AgeRange
	Yes	Caffè latte	Sometimes	Female	Lebanon	18 to 29
	Yes	Americano	Yes	Female	Lebanon	30 to 39
	No	Nescafe	Yes	Female	Lebanon	18 to 29

File Loading

The result is a link. For example:

https://github.com/andvise/DataAnalyticsDatasets/blob/2d91808c5f66cc4bc54ebe6cd04b1db47b3aec8a/coffee.csv

To make it directly readable by pandas add”:

`?raw=true`

https://github.com/andvise/DataAnalyticsDatasets/blob/2d91808c5f66cc4bc54ebe6cd04b1db47b3aec8a/coffee.csv?raw=true

File Loading

```
import pandas as pd
path =
"https://github.com/andvise/DataAnalyticsDatasets/blob/2d91808c5f66cc
4bc54ebe6cd04b1db47b3aec8a/coffee.csv?raw=true"
df = pd.read_csv(path)
df
```

File Loading Errors

FileNotFoundError: File b'filename.csv' does not exist

A File Not Found error is typically an issue with path setup, current directory, or file name confusion (file extension can play a part here!)

Read parameters

```
pandas.read_csv(filepath_or_buffer, sep=NoDefault.no_default, delimiter=None,
header='infer', names=NoDefault.no_default, index_col=None, usecols=None,
squeeze=False, prefix=NoDefault.no_default, mangle_dupe_cols=True, dtype=None,
engine=None, converters=None, true_values=None, false_values=None,
skipinitialspace=False, skiprows=None, skipfooter=0, nrows=None, na_values=None,
keep_default_na=True, na_filter=True, verbose=False, skip_blank_lines=True,
parse_dates=False, infer_datetime_format=False, keep_date_col=False,
date_parser=None, dayfirst=False, cache_dates=True, iterator=False, chunksize=None,
compression='infer', thousands=None, decimal='.', lineterminator=None, quotechar='"',
quoting=0, doublequote=True, escapechar=None, comment=None, encoding=None,
encoding_errors='strict', dialect=None, error_bad_lines=None, warn_bad_lines=None,
on_bad_lines=None, delim_whitespace=False, low_memory=True, memory_map=False,
float_precision=None, storage_options=None)
```

Read parameters

Read CSV has many parameters that can be used to adapt its usage to your file.

For most of them the defaults values are fine.

For more info check:

https://pandas.pydata.org/docs/reference/api/pandas.read_csv.html

File Writing

Pandas can be use to save data into external files.

- This is really useful to share them or to visualize them in Excel.
- Or to save data during different sessions.

Different formats read and write

Data Formate	Read	Save
csv	`pd.read_csv()`	`df.to_csv()`
json	`pd.read_json()`	`df.to_json()`
excel	`pd.read_excel()`	`df.to_excel()`
hdf	`pd.read_hdf()`	`df.to_hdf()`
sql	`pd.read_sql()`	`df.to_sql()`
...	...	...

DataFrames

Titanic Dataset

The original Titanic dataset, describing the survival status of individual passengers on the Titanic.

- Available on Github with description

DataFrame - Shape

DataFrame.shape

- returns the number of rows and columns.

Data Frame - Shape

main.py ×

```
import pandas as pd

# Import the data from the csv file
df = pd.read_csv("titanic.csv")

# Display the size of the dataset
print(df.shape)
```

Console Shell

```
(887, 9)
```

Number of cols

Number of rows

Data Frame – Data Types

main.py ×

```python
import pandas as pd

# Import the data from the csv file
df = pd.read_csv("titanic.csv")

# Data types of all columns
print(df.dtypes)
```

Console Shell

```
PassengerId                  int64
Survived                     int64
Pclass                       int64
Name                        object
Sex                         object
Age                        float64
Siblings/Spouses Aboard      int64
Parents/Children Aboard    float64
Fare                       float64
dtype: object
```

Data Frame – Data Types

Pandas Type	Native Python Type	Description
object	string or mixed	The most general dtype. Will be assigned to your column if column has mixed types (numbers and strings).
int64	int	Numeric characters. 64 refers to the memory allocated to hold this character.
float64	float	Numeric characters with decimals. If a column contains numbers and NaNs(see below), pandas will default to float64, in case your missing value has a decimal.
bool	bool	True/False values
datetime64, timedelta[ns]	N/A (but see the datetime module in Python's standard library)	Values meant to hold time data. Look into these for time series experiments.
category	N/A	Finite list of text values

Data Frame – Data Types

main.py ×

```
import pandas as pd

# Import the data from the csv file
df = pd.read_csv("titanic.csv")

# Data type of a single column
print(df['Age'].dtype)
```

Console Shell

```
float64
```

Data Frames attributes

Python objects have **attributes** and **methods**.

- The **attributes** are variables stored in that object
  - To access them: `object.attribute`

- The **methods** are functions of that object
  - To access them: `object.method()`

Data Frames attributes

df.attribute	Description
dtypes	list the types of the columns
columns	list the column names
axes	list the row labels and column names
ndim	number of dimensions
size	number of elements
shape	return a tuple representing the dimensionality
values	numpy representation of the data

Data Frames attributes

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Hands-on exercises

- ✓Find how many records this data frame has;
- ✓How many elements are there?
- ✓What are the column names?
- ✓What types of columns we have in this data frame?

That's all Folks!
SUPERB wallpapers